

Create API endpoint GET /api/users with that response all users id and name

http://localhost:8080/api/user/users

Save

No Environment

GEThttp://localhost:8080/api/user/usersSend

ParamsAuthorizationHeaders (7)BodyPre-request ScriptTestsSettingsCodeCookies

Query Params

| Key | Value |
|-----|-------|
| Key | Value |

BodyCookiesHeaders (5)Test Results

Status: 200 OKTime: 285 msSize: 1.24 KB

PrettyRawPreviewJSON

```
1 [{"id":1,"firstName":"Nghia","lastName":"Nguyen","otherName":"Trung","gender":null,"address":"123 Linh Xuan","sateOfOrigin":null,"accountNumber":"2024786946","accountBalance":0.00,"email":"ntnghia@hcmiu.edu.vn","phoneNumber":"123456789","alternativePhoneNumber":"0763503524","status":"ACTIVE","createAt":"2024-11-28T09:33:04.227349","modifiedAt":"2024-11-28T09:33:04.227349"}, {"id":2,"firstName":"Vu","lastName":"Nhan","otherName":"Thanh","gender":null,"address":"536 Au Co","sateOfOrigin":null,"accountNumber":"2024891097","accountBalance":0.00,"email":"vuthanhnhhan0105@gmail.com","phoneNumber":"123456789","alternativePhoneNumber":"0981787077","status":"ACTIVE","createAt":"2024-11-28T13:45:24.191555","modifiedAt":"2024-11-28T13:45:24.191555"}, {"id":3,"firstName":"Tom","lastName":"Jerry","otherName":"Spike","gender":null,"address":"MGM","sateOfOrigin":null,"accountNumber":"2024650973","accountBalance":0.00,"email":"Tom&Jerry@gmail.com","phoneNumber":"123456789","alternativePhoneNumber":"987654321","status":null,"createAt":"2024-11-28T13:46:06.406719","modifiedAt":"2024-11-28T14:02:46.315395"}]
```

Create API endpoint GET /api/user/{id} which get detail of a specific user.

http://localhost:8080/api/user/1

Save

No Environment

GEThttp://localhost:8080/api/user/1Send

ParamsAuthorizationHeaders (7)BodyPre-request ScriptTestsSettingsCodeCookies

Query Params

| Key | Value |
|-----|-------|
| Key | Value |

BodyCookiesHeaders (5)Test Results

Status: 200 OKTime: 16 msSize: 538 B

PrettyRawPreviewJSON

```
1 [{"id":1,"firstName":"Nghia","lastName":"Nguyen","otherName":"Trung","gender":null,"address":"123 Linh Xuan","sateOfOrigin":null,"accountNumber":"2024786946","accountBalance":0.00,"email":"ntnghia@hcmiu.edu.vn","phoneNumber":"123456789","alternativePhoneNumber":"0763503524","status":"ACTIVE","createAt":"2024-11-28T09:33:04.227349","modifiedAt":"2024-11-28T09:33:04.227349"}]
```

Create API endpoint PUT /api/users{id} which update a specific user

| ier | create_at                  | email                      | first_name | gender | last_name | modified_at                | other_name | phone_number | sate_of_origin | status |
|-----|----------------------------|----------------------------|------------|--------|-----------|----------------------------|------------|--------------|----------------|--------|
|     | 2024-11-28 09:33:04.227349 | ntnghia@hcmiu.edu.vn       | Nghia      | NULL   | Nguyen    | 2024-11-28 09:33:04.227349 | Trung      | 123456789    | NULL           | ACTIVE |
|     | 2024-11-28 13:45:24.191555 | vuthanhnhhan0105@gmail.com | Vu         | NULL   | Nhan      | 2024-11-28 13:45:24.191555 | Thanh      | 123456789    | NULL           | ACTIVE |
|     | 2024-11-28 13:46:06.406719 | Tom&Jerry@gmail.com        | Tom        | NULL   | Jerry     | 2024-11-28 14:02:46.315395 | Spike      | 123456789    | NULL           | NULL   |

HTTP <http://localhost:8080/api/user/3> Save No Environment

PUT <http://localhost:8080/api/user/3> Send

Params Authorization Headers (9) **Body** Pre-request Script Tests Settings Code Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL Text

```

1 {
2   "firstName": "Tom",
3   "lastName": "Tom",
4   "otherName": "Tom",
5   "gender": "male",
6   "address": "123 Linh Xuan",
7   "stateOforigin": "HCM",
8   "accountBalance": 2000,
9   "email": "Tom@hcmiu.edu.vn",
10  "phoneNumber": "123456789",
11  "alternativePhoneNumber": "987654321"
12 }

```

| id | create_at                  | email                     | first_name | gender | last_name | modified_at                | other_name | phone_number | state_of_origin | status |
|----|----------------------------|---------------------------|------------|--------|-----------|----------------------------|------------|--------------|-----------------|--------|
| 1  | 2024-11-28 09:33:04.227349 | ntnghia@hcmiu.edu.vn      | Nghia      | NULL   | Nguyen    | 2024-11-28 09:33:04.227349 | Trung      | 123456789    | NULL            | ACTIVE |
| 2  | 2024-11-28 13:45:24.191555 | vuthanhnhan0105@gmail.com | Vu         | NULL   | Nhan      | 2024-11-28 13:45:24.191555 | Thanh      | 123456789    | NULL            | ACTIVE |
| 3  | 2024-11-28 13:46:06.406719 | Tom@hcmiu.edu.vn          | Tom        | NULL   | Tom       | 2024-11-28 18:23:33.629807 | Tom        | 123456789    | NULL            | ACTIVE |

Create API endpoint DELETE /api/user/{id} which delete specific user

HTTP <http://localhost:8080/api/user/3> Save No Environment

DELETE <http://localhost:8080/api/user/3> Send

Params Authorization Headers (9) **Body** Pre-request Script Tests Settings Code Cookies

Query Params

| Key | Value |
|-----|-------|
| Key | Value |

Body Cookies Headers (5) Test Results

Status: 200 OK Time: 29 ms Size: 209 B

| id | create_at                  | email                     | first_name | gender | last_name | modified_at                | other_name | phone_number | state_of_origin | status |
|----|----------------------------|---------------------------|------------|--------|-----------|----------------------------|------------|--------------|-----------------|--------|
| 1  | 2024-11-28 09:33:04.227349 | ntnghia@hcmiu.edu.vn      | Nghia      | NULL   | Nguyen    | 2024-11-28 09:33:04.227349 | Trung      | 123456789    | NULL            | ACTIVE |
| 2  | 2024-11-28 13:45:24.191555 | vuthanhnhan0105@gmail.com | Vu         | NULL   | Nhan      | 2024-11-28 13:45:24.191555 | Thanh      | 123456789    | NULL            | ACTIVE |
| 3  | 2024-11-28 13:46:06.406719 | Tom@hcmiu.edu.vn          | Tom        | NULL   | Tom       | 2024-11-28 18:23:33.629807 | Tom        | 123456789    | NULL            | ACTIVE |

UserController.java:

package com.wadlab.academy\_bank.controller;

import java.util.List;

```
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.DeleteMapping;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.PutMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
```

```
import com.wadlab.academy_bank.dto.BankResponse;
import com.wadlab.academy_bank.dto.UserRequest;
import com.wadlab.academy_bank.entity.User;
import com.wadlab.academy_bank.services.impl.UserService;
```

```
@RestController
```

```
@RequestMapping("/api/user")
```

```
public class UserController {
```

```
    @Autowired
```

```
    UserService userService;
```

```
    @PostMapping
```

```
    public BankResponse createAccount(@RequestBody UserRequest userRequest) {
        return userService.createAccount(userRequest);
    }
```

```
    @GetMapping("/users")
```

```
    public List<User> getAllUsers() {
```

```
    return userService.getAllUsers();  
}
```

```
@GetMapping("/{id}")  
public User getUserById(@PathVariable Long id) {  
    return userService.getUserById(id);  
}
```

```
@DeleteMapping("/{id}")  
public String deleteUser(@PathVariable Long id) {  
    userService.deleteUser(id);  
    return "User with ID " + id + " has been deleted successfully.";  
}
```

```
@PutMapping("/{id}")  
public User updateUser(@PathVariable Long id, @RequestBody UserRequest userRequest) {  
    return userService.updateUser(id, userRequest);  
}  
}
```

AccountInfo.java:

```
package com.wadlab.academy_bank.dto;
```

```
import lombok.AllArgsConstructor;  
import lombok.Builder;  
import lombok.Data;  
import lombok.NoArgsConstructor;
```

```
import java.math.BigDecimal;
```

@Data

@Builder

@NoArgsConstructor

@AllArgsConstructor

```
public class AccountInfo {  
    private String accountName;  
    private BigDecimal accountBalance;  
    private String accountNumber;  
}
```

BankResponse.java:

```
package com.wadlab.academy_bank.dto;
```

```
import lombok.*;
```

@Data

@Builder

@NoArgsConstructor

@AllArgsConstructor

```
public class BankResponse {  
    private String responseCode;  
    private String responseMessage;  
    private AccountInfo AccountInfo;  
}
```

UserRequest.java:

```
package com.wadlab.academy_bank.dto;
```

```
import lombok.*;
```

```
@Data
```

```
@Builder
```

```
@AllArgsConstructor
```

```
@NoArgsConstructor
```

```
public class UserRequest {
```

```
    private String firstName;
```

```
    private String lastName;
```

```
    private String otherName;
```

```
    private String gender;
```

```
    private String address;
```

```
    private String sateOfOrigin;
```

```
    private String accountNumber;
```

```
    private String email;
```

```
    private String phoneNumber;
```

```
    private String alternativePhoneNumber;
```

```
    private String status;
```

```
}
```

User.java:

```
package com.wadlab.academy_bank.entity;
```

```
import java.math.BigDecimal;
```

```
import java.time.LocalDateTime;
```

```
import org.hibernate.annotations.CreationTimestamp;
```

```
import org.hibernate.annotations.UpdateTimestamp;
```

```
import jakarta.persistence.*;
```

```
import lombok.*;
```

```
@Getter
```

```
@Setter
```

```
@NoArgsConstructor
```

```
@AllArgsConstructor
```

```
@Builder
```

```
@Entity
```

```
@Table(name = "users")
```

```
public class User {
```

```
    @Id
```

```
    @GeneratedValue(strategy = GenerationType.IDENTITY)
```

```
    private Long id;
```

```
    private String firstName;
```

```
    private String lastName;
```

```
    private String otherName;
```

```
    private String gender;
```

```
    private String address;
```

```
    private String sateOfOrigin;
```

```
    private String accountNumber;
```

```
    private BigDecimal accountBalance;
```

```
    private String email;
```

```
    private String phoneNumber;
```

```
    private String alternativePhoneNumber;
```

```
    private String status;
```

```
    @CreationTimestamp
```

```
private LocalDateTime createdAt;  
@UpdateTimestamp  
private LocalDateTime modifiedAt;  
}
```

UserRepository.java:

```
package com.wadlab.academy_bank.repository;  
  
import org.springframework.data.jpa.repository.JpaRepository;  
  
import com.wadlab.academy_bank.entity.User;  
  
public interface UserRepository extends JpaRepository<User, Long>{  
    Boolean existsByEmail(String email);  
  
}
```

UserService.java:

```
package com.wadlab.academy_bank.services.impl;  
  
import java.util.List;  
  
import com.wadlab.academy_bank.dto.BankResponse;  
import com.wadlab.academy_bank.dto.UserRequest;  
import com.wadlab.academy_bank.entity.User;  
  
public interface UserService {  
    BankResponse createAccount(UserRequest userRequest);
```

```
List<User> getAllUsers();

User getUserById(Long id);

void deleteUser(Long id);

User updateUser(Long id, UserRequest userRequest);
}
```

UserServiceImpl.java:

```
package com.wadlab.academy_bank.services.impl;

import java.math.BigDecimal;
import java.util.List;
import java.util.Optional;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import com.wadlab.academy_bank.dto.AccountInfo;
import com.wadlab.academy_bank.dto.BankResponse;
import com.wadlab.academy_bank.dto.UserRequest;
import com.wadlab.academy_bank.entity.User;
import com.wadlab.academy_bank.repository.UserRepository;
import com.wadlab.academy_bank.utils.AccountUtilis;
```

@Service

```
public class UserServiceImpl implements UserService {
```

@Autowired

```
UserRepository userRepository;
```

@Override

```
public BankResponse createAccount(UserRequest userRequest) {
```

```
    if(userRepository.existsByEmail(userRequest.getEmail())) {
```

```
        return BankResponse.builder()
```

```
            .responseCode(AccountUtilis.ACCOUNT_EXISTS_CODE)
```

```
            .responseMessage(AccountUtilis.ACCOUNT_EXISTS_MESSAGE)
```

```
            .AccountInfo(null)
```

```
            .build();
```

```
    }
```

```
    User newUser = User.builder()
```

```
        .firstName(userRequest.getFirstName())
```

```
        .lastName(userRequest.getLastName())
```

```
        .otherName(userRequest.getOtherName())
```

```
        .address(userRequest.getAddress())
```

```
        .sateOfOrigin(userRequest.getSateOfOrigin())
```

```
        .accountNumber(AccountUtilis.generateAccountNumber())
```

```
        .accountBalance(BigDecimal.ZERO)
```

```
        .phoneNumber(userRequest.getPhoneNumber())
```

```
        .alternativePhoneNumber(userRequest.getAlternativePhoneNumber())
```

```
        .email(userRequest.getEmail())
```

```
        .status("ACTIVE")
```

```
        .build();
```

```

        User savedUser = userRepository.save(newUser);

        AccountInfo accountInfo = AccountInfo.builder()

            .accountBalance(savedUser.getAccountBalance())

            .accountNumber(savedUser.getAccountNumber())

            .accountName(savedUser.getFirstName() + " " + savedUser.getLastName() + " " +
savedUser.getOtherName())

            .build();

        return BankResponse.builder()

            .responseCode(AccountUtilis.ACCOUNT_CREATION_SUCCESS_CODE)

            .responseMessage(AccountUtilis.ACCOUNT_CREATION_SUCCESS_MESSAGE)

            .AccountInfo(accountInfo)

            .build();
    }

```

@Override

```

public List<User> getAllUsers() {
    return userRepository.findAll();
}

```

@Override

```

public User getUserById(Long id) {
    Optional<User> user = userRepository.findById(id);
    return user.orElse(null);
}

```

@Override

```

public void deleteUser(Long id) {

```

```

Optional<User> user = userRepository.findById(id);
if (user.isPresent()) {
    userRepository.deleteById(id);
} else {
    throw new RuntimeException("User not found with ID: " + id);
}
}

```

@Override

```

public User updateUser(Long id, UserRequest userRequest) {
    Optional<User> existingUser = userRepository.findById(id);
    if (existingUser.isPresent()) {
        User userToUpdate = existingUser.get();
        // Update user properties
        userToUpdate.setFirstName(userRequest.getFirstName());
        userToUpdate.setLastName(userRequest.getLastName());
        userToUpdate.setOtherName(userRequest.getOtherName());
        userToUpdate.setAddress(userRequest.getAddress());
        userToUpdate.setSateOfOrigin(userRequest.getSateOfOrigin());
        userToUpdate.setPhoneNumber(userRequest.getPhoneNumber());
        userToUpdate.setAlternativePhoneNumber(userRequest.getAlternativePhoneNumber());
        userToUpdate.setEmail(userRequest.getEmail());
        userToUpdate.setStatus(userRequest.getStatus());

        return userRepository.save(userToUpdate);
    } else {
        throw new RuntimeException("User not found with ID: " + id);
    }
}

```

```
}
```

AccountUtilis.java:

```
package com.wadlab.academy_bank.utils;
```

```
import java.time.Year;
```

```
public class AccountUtilis {
```

```
    public static final String ACCOUNT_EXISTS_CODE = "001";
```

```
    public static final String ACCOUNT_EXISTS_MESSAGE = "This user already exists!";
```

```
    public static final String ACCOUNT_CREATION_SUCCESS_CODE = "002";
```

```
    public static final String ACCOUNT_CREATION_SUCCESS_MESSAGE = "Account has been  
successfully created";
```

```
    public static String generateAccountNumber() {
```

```
        // 2023 + randonSixDigits
```

```
        Year currentYear = Year.now();
```

```
        int min = 100000;
```

```
        int max = 999999;
```

```
        int randNumber = (int) Math.floor(Math.random() * (max - min + 1));
```

```
        String year = String.valueOf(currentYear);
```

```
        String randomNumber = String.valueOf(randNumber);
```

```
        StringBuilder accountNumber = new StringBuilder();
```

```
        return accountNumber
            .append(year)
            .append(randomNumber)
            .toString();
    }

}
```

AcademyBankApplication.java:

```
package com.wadlab.academy_bank;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class AcademyBankApplication {

    public static void main(String[] args) {
        SpringApplication.run(AcademyBankApplication.class, args);
    }

}
```

AcademyBankApplicationTests.java:

```
package com.wadlab.academy_bank;

import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;
```

```
@SpringBootTest
class AcademyBankApplicationTests {

    @Test
    void contextLoads() {
    }

}
```